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**Java SE 27 & JDK 27**

# Class Attributes

java.lang.Object
    java.util.jar.Attributes

**All Implemented Interfaces:**

Cloneable, Map<Object,Object>

---

```
public class Attributes
extends Object
implements Map<Object,Object>, Cloneable
```

The Attributes class maps Manifest attribute names to associated string values. Valid attribute names are case-insensitive, are restricted to the ASCII characters in the set [0-9a-zA-Z_-], and cannot exceed 70 characters in length. There must be a colon and a SPACE after the name; the combined length will not exceed 72 characters. Attribute values can contain any characters and will be UTF8-encoded when written to the output stream. See the JAR File Specification for more information about valid attribute names and values.

This map and its views have a predictable iteration order, namely the order that keys were inserted into the map, as with LinkedHashMap.

**Since:**

1.2

**External Specifications**

JAR File Specification

**See Also:**

Manifest

## *Nested Class Summary*

**Nested Classes**

| Modifier and Type | Class | Description |
|---|---|---|
| static class | **Attributes.Name** | The Attributes.Name class represents an attribute name stored in this Map. |

**Nested classes/interfaces declared in interface Map**

Map.Entry<K,V>

## Field Summary

**Fields**

| Modifier and Type | Field | Description |
|---|---|---|
| protected Map<Object,Object> | map | The attribute name-value mappings. |

## Constructor Summary

**Constructors**

| Constructor | Description |
|---|---|
| Attributes() | Constructs a new, empty Attributes object with default size. |
| Attributes(int size) | Constructs a new, empty Attributes object with the specified initial size. |
| Attributes(Attributes attr) | Constructs a new Attributes object with the same attribute name-value mappings as in the specified Attributes. |

## Method Summary

**All Methods** | **Instance Methods**

| Modifier and Type | Method | Description |
|---|---|---|
| void | clear() | Removes all attributes from this Map. |
| Object | clone() | Returns a copy of the Attributes, implemented as follows: |
| boolean | containsKey(Object name) | Returns true if this Map contains the specified attribute name (key). |
| boolean | containsValue(Object value) | Returns true if this Map maps one or more attribute names (keys) to the specified value. |

| | | |
|---|---|---|
| Set<Map.Entry<Object, Object>> | **entrySet**() | Returns a Collection view of the attribute name-value mappings contained in this Map. |
| boolean | **equals**(Object o) | Compares the specified object to the underlying map for equality. |
| Object | **get**(Object name) | Returns the value of the specified attribute name, or null if the attribute name was not found. |
| String | **getValue**(String name) | Returns the value of the specified attribute name, specified as a string, or null if the attribute was not found. |
| String | **getValue**(Attributes.Name name) | Returns the value of the specified Attributes.Name, or null if the attribute was not found. |
| int | **hashCode**() | Returns the hash code value for this Map. |
| boolean | **isEmpty**() | Returns true if this Map contains no attributes. |
| Set<Object> | **keySet**() | Returns a Set view of the attribute names (keys) contained in this Map. |
| Object | **put**(Object name, Object value) | Associates the specified value with the specified attribute name (key) in this Map. |
| void | **putAll**(Map<?,?> attr) | |

| | | |
|---|---|---|
| | | Copies all of the attribute name-value mappings from the specified Attributes to this Map. |
| String | **putValue**(String name, String value) | Associates the specified value with the specified attribute name, specified as a String. |
| Object | **remove**(Object name) | Removes the attribute with the specified name (key) from this Map. |
| int | **size**() | Returns the number of attributes in this Map. |
| Collection<Object> | **values**() | Returns a Collection view of the attribute values contained in this Map. |

**Methods declared in class Object**

finalize, getClass, notify, notifyAll, toString, wait, wait, wait

**Methods declared in interface Map**

compute, computeIfAbsent, computeIfPresent, forEach, getOrDefault, merge, putIfAbsent, remove, replace, replace, replaceAll

## *Field Details*

### map

```
protected Map<Object,Object> map
```

The attribute name-value mappings.

## *Constructor Details*

### Attributes

```
public Attributes()
```

Constructs a new, empty Attributes object with default size.

### Attributes

```
public Attributes(int size)
```

Constructs a new, empty Attributes object with the specified initial size.

**Parameters:**

`size` - the initial number of attributes

### Attributes

```
public Attributes(Attributes attr)
```

Constructs a new Attributes object with the same attribute name-value mappings as in the specified Attributes.

**Parameters:**

`attr` - the specified Attributes

## *Method Details*

### get

```
public Object get(Object name)
```

Returns the value of the specified attribute name, or null if the attribute name was not found.

**Specified by:**

`get` in interface `Map<Object,Object>`

**Parameters:**

`name` - the attribute name

**Returns:**

the value of the specified attribute name, or null if not found.

### getValue

```
public String getValue(String name)
```

Returns the value of the specified attribute name, specified as a string, or null if the attribute was not found. The attribute name is case-insensitive.

This method is defined as:

```
    return (String)get(new Attributes.Name((String)name));
```

**Parameters:**

`name` - the attribute name as a string

**Returns:**

the String value of the specified attribute name, or null if not found.

**Throws:**

`IllegalArgumentException` - if the attribute name is invalid

## getValue

```
public String getValue(Attributes.Name name)
```

Returns the value of the specified Attributes.Name, or null if the attribute was not found.

This method is defined as:

```
    return (String)get(name);
```

**Parameters:**

`name` - the Attributes.Name object

**Returns:**

the String value of the specified Attribute.Name, or null if not found.

## put

```
public Object put(Object name,
                  Object value)
```

Associates the specified value with the specified attribute name (key) in this Map. If the Map previously contained a mapping for the attribute name, the old value is replaced.

**Specified by:**

`put` in interface `Map<Object,Object>`

**Parameters:**

`name` - the attribute name

`value` - the attribute value

**Returns:**

the previous value of the attribute, or null if none

**Throws:**

`ClassCastException` - if the name is not a Attributes.Name or the value is not a String

## putValue

```
public String putValue(String name,
                       String value)
```

Associates the specified value with the specified attribute name, specified as a String. The attributes name is case-insensitive. If the Map previously contained a mapping for the attribute name, the old value is replaced.

This method is defined as:

```
return (String)put(new Attributes.Name(name), value);
```

**Parameters:**

`name` - the attribute name as a string

`value` - the attribute value

**Returns:**

the previous value of the attribute, or null if none

**Throws:**

`IllegalArgumentException` - if the attribute name is invalid

## remove

```
public Object remove(Object name)
```

Removes the attribute with the specified name (key) from this Map. Returns the previous attribute value, or null if none.

**Specified by:**

`remove` in interface `Map<Object,Object>`

**Parameters:**

`name` - attribute name

**Returns:**

the previous value of the attribute, or null if none

## containsValue

```
public boolean containsValue(Object value)
```

Returns true if this Map maps one or more attribute names (keys) to the specified value.

**Specified by:**

`containsValue` in interface `Map<Object,Object>`

**Parameters:**

`value` - the attribute value

**Returns:**

true if this Map maps one or more attribute names to the specified value

## containsKey

```
public boolean containsKey(Object name)
```

Returns true if this Map contains the specified attribute name (key).

**Specified by:**

`containsKey` in interface `Map<Object,Object>`

**Parameters:**

`name` - the attribute name

**Returns:**

true if this Map contains the specified attribute name

## putAll

```
public void putAll(Map<?,?> attr)
```

Copies all of the attribute name-value mappings from the specified Attributes to this Map. Duplicate mappings will be replaced.

**Specified by:**

`putAll` in interface `Map<Object,Object>`

**Parameters:**

`attr` - the Attributes to be stored in this map

**Throws:**

`ClassCastException` - if attr is not an Attributes

## clear

```
public void clear()
```

Removes all attributes from this Map.

**Specified by:**

`clear` in interface `Map<Object,Object>`

## size

```
public int size()
```

Returns the number of attributes in this Map.

**Specified by:**

`size` in interface `Map<Object,Object>`

**Returns:**

the number of key-value mappings in this map

## isEmpty

```
public boolean isEmpty()
```

Returns true if this Map contains no attributes.

**Specified by:**

`isEmpty` in interface `Map<Object,Object>`

**Returns:**

`true` if this map contains no key-value mappings

## keySet

```
public Set<Object> keySet()
```

Returns a Set view of the attribute names (keys) contained in this Map.

**Specified by:**

`keySet` in interface `Map<Object,Object>`

**Returns:**

a set view of the keys contained in this map

## values

```
public Collection<Object> values()
```

Returns a Collection view of the attribute values contained in this Map.

**Specified by:**

`values` in interface `Map<Object,Object>`

**Returns:**

a collection view of the values contained in this map

## entrySet

```
public Set<Map.Entry<Object,Object>> entrySet()
```

Returns a Collection view of the attribute name-value mappings contained in this Map.

**Specified by:**

`entrySet` in interface `Map<Object,Object>`

**Returns:**

a set view of the mappings contained in this map

## equals

```
public boolean equals(Object o)
```

Compares the specified object to the underlying map for equality. Returns true if the given object is also a Map and the two maps represent the same mappings.

**Specified by:**

equals in interface Map<Object,Object>

**Overrides:**

equals in class Object

**Parameters:**

o - the Object to be compared

**Returns:**

true if the specified Object is equal to this Map

**See Also:**

Object.hashCode(), HashMap

## hashCode

```
public int hashCode()
```

Returns the hash code value for this Map.

**Specified by:**

hashCode in interface Map<Object,Object>

**Overrides:**

hashCode in class Object

**Returns:**

the hash code value for this map

**See Also:**

Map.Entry.hashCode(), Object.equals(Object), Map.equals(Object)

## clone

```
public Object clone()
```

Returns a copy of the Attributes, implemented as follows:

```
public Object clone() { return new Attributes(this); }
```

Since the attribute names and values are themselves immutable, the Attributes returned can be safely modified without affecting the original.

**Overrides:**

`clone` in class `Object`

**Returns:**

a clone of this instance.

**See Also:**

`Cloneable`

---

Report a bug or suggest an enhancement
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